

NEET → 18. (4 marks) PW

JKBOSE

🔥 2026 CLASS 11th 🔥

THE LIVING WORLD

BOTANY LECTURE-1

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TOPICS TO BE COVERED

- 1 Biodiversity ✓
- 2 Need for Classification ✓
- 3 Taxonomy ✓
- 4 Systematics ✓

Botany → class^{11th} (BAAZ)*

↓
Chapters:

- ①. The living world.*
- ②. Biological Classification.
- ③. Plant Kingdom
- ④. Morphology (Angiosperms)
- ⑤. Anatomy (Angiosperms)
- ⑥. Photosynthesis
- ⑦. Respiration.
- ⑧. Plant growth & Development

BIODIVERSITY

* $\begin{cases} \text{Bio} \longrightarrow \text{life / living.} \\ \text{diversity} \longrightarrow \text{forms / variety.} \end{cases}$

↳ Biodiversity — It is total sum of all the living organisms present on planet Earth.

↳ 1.7 — 1.8 million Species present on Earth } taxonomically known sps.
 ↓
 1.2 million — Animal sps.
 0.5 million — Plant sps.

NEED FOR CLASSIFICATION

- There is huge no. of plant and animal species which vary greatly in size, form, str., complexity etc.
- It is impossible to study all of them individually.
- Thus - to study them - they are classified on the basis of easily observable characters into different categories.

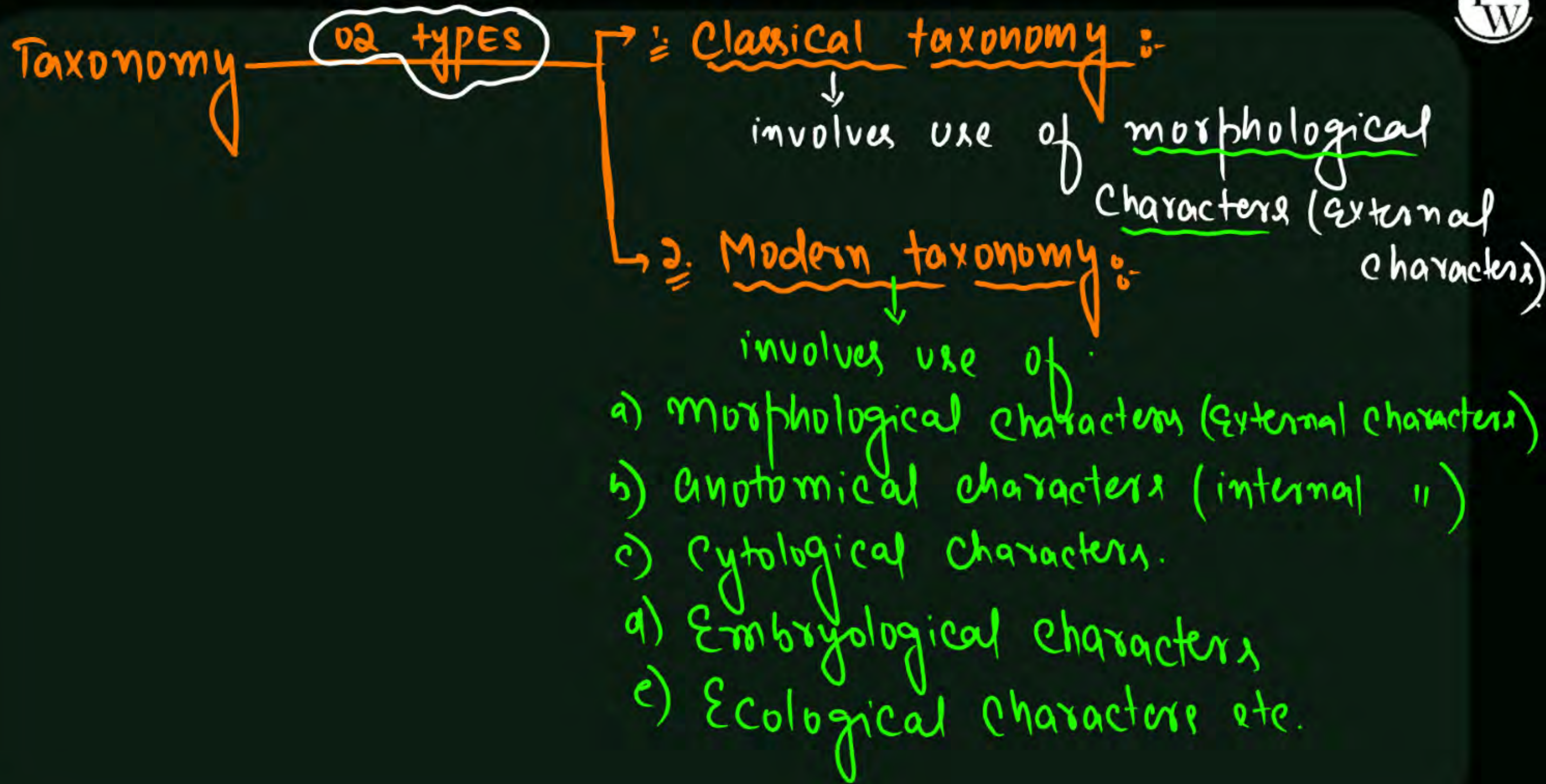
TAXONOMY

≡ $\left\{ \begin{array}{l} \text{Taxis} \text{ ————— arrangement} \\ \text{NOMO's} \text{ ————— rules / laws} \end{array} \right.$

↳ Taxonomy — branch of science which deals with classification of living organisms (by following rules).

↳ Term taxonomy — Coined by "A. P. deCandolle"

↳ Father of taxonomy — "Carolus Linnaeus"



Taxonomy → involves following steps:

- ✓ ①. Characterization.
- ✓ ②. Identification.
- ✓ ③. Classification.
- ✓ ④. Nomenclature.

*

Note:-

→ Taxonomy = Characterization ✓
+ Identification ✓
+ Classification ✓
+ Nomenclature ✓

TAXONOMY INVOLVES FOLLOWING STEPS

① CHARACTERISATION:

- Organism is described according to its all morphological, anatomical, physiological and behavioural properties.

② IDENTIFICATION:

- Process by which an organism is recognized from the other already known organisms and is assigned to a particular taxonomic group.

③ CLASSIFICATION:

- Process by which an organism is grouped into convenient categories on the basis of some easily observable characters.

④ NOMENCLATURE:

- It is the naming of an organism according to established rules.

SYSTEMATICS

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derived from latin word — "Systema"
↓ means

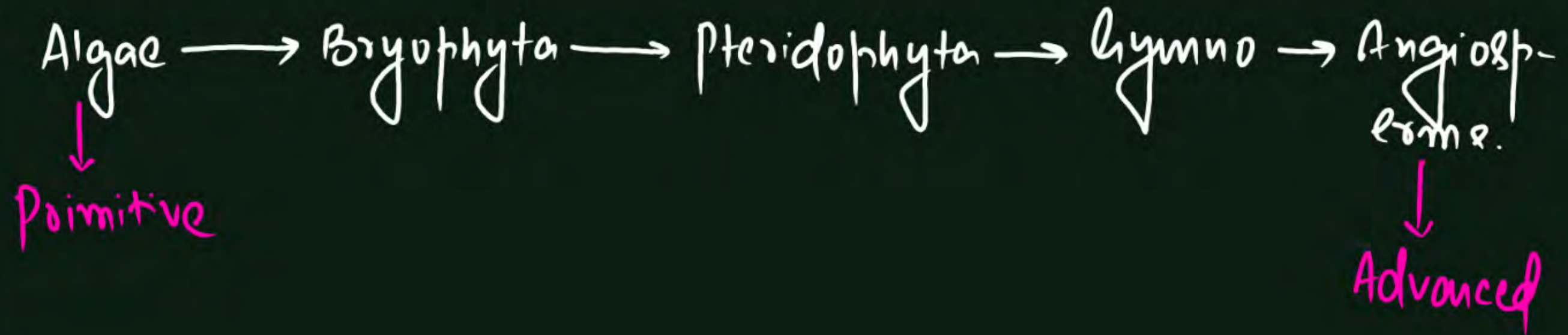
↳ Term "Systematics" — used by "Systematic arrangement"
C. Linnaeus in his book "Systema naturae"

→ Systematics involves $\left\{ \begin{array}{l} \text{diversity among organisms.} \\ \text{and} \\ \text{phylogeny (Evolutionary relationship} \\ \text{among organisms).} \end{array} \right.$

*

* Note:

Systematics = Taxonomy + Phylogeny



QUESTION

The process by which any thing is grouped into convenient categories based on some easily observable characters is called:

1 Identification

2 Systematics

✓ 3 Classification

4 Nomenclature



QUESTION

Systematics was first time proposed by

1

C. Linnaeus in Pinax

2

Casper Bauhin in Pinax

3

Casper Bauhin in Species Plantarum

✓ 4

C. Linnaeus in Systema Naturae



QUESTION

Who wrote systema naturae:

- 1 ✓ Linnaeus
- 2 Mayr
- 3 John Ray
- 4 De Candolle



Thank
You